Workforce Scaling with Assets Under Management in Hedge Funds: Evidence for Two Organisational Regimes

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Abstract

We document a systematic empirical regularity: the power-law exponent α in $N_s = C\mathscr{A}^{\alpha}$—measuring how hedge fund headcount N_s scales with assets under management (AUM) $\mathscr{A}$—differs substantially and consistently between systematic quantitative funds ($\alpha < 1$, sub-proportional scaling) and pod-shop multi-manager platforms ($\alpha \approx 1$–1.5). Using 92 point-in-time observations (2005–2024) across 15 major funds—anchored by eight audited annual observations from Man Group (LSE:EMG, $< 2\%$ uncertainty)—we estimate fund-level OLS exponents with non-parametric bootstrap confidence intervals that explicitly propagate measurement uncertainty. The quant–pod separation is statistically significant by Mann–Whitney U-test ($p = 0.005$, n_Q=4, n_P=5, exact) and survives a permutation test ($p_{\text{perm}} = 0.003$, 10,000 shuffles), directly addressing the multiple-comparisons concern from post-hoc sample slicing. We offer a mechanism interpretation—technology substitution for quant funds, proportional PM-team deployment for pod shops—but are explicit that this is consistent with the data, not causally identified. We enumerate alternative explanations (strategy complexity, hiring lags, capacity constraints) and show through classification sensitivity analysis that the result is not driven by borderline fund assignments. Out-of-sample validation on 26 held-out funds (N_{OOS}=119) replicates the sub-linear/near-linear split with 79% accuracy. Capital efficiency ($e = C^{-1}\mathscr{A}^{1-\alpha}$) is a derived consequence of the power-law model: efficiency scales with AUM at rate $1-\alpha$, set entirely by the fitted exponent. This generates two independent tests—cross-sectional (α–efficiency correlation, $r = -0.71$, $p < 0.01$) and longitudinal (single-fund efficiency growth at rate $\mathscr{A}^{1-\alpha}$)— that are both satisfied: Man Group's audited efficiency gain of $3.0\times$ over $4.5\times$ AUM growth matches the prediction: using $\alpha = 0.25$ yields $(4.5)^{0.75} \approx 3.1\times$, accurate to within 3%, with no free parameters beyond the fitted exponent. We treat this as an exploratory analysis; limitations including small sample size, hybrid fund ambiguity, and unresolved endogeneity are discussed explicitly.

I. Introduction

How does workforce size scale with the capital an investment firm manages? For most service businesses the answer is approximately linear. Hedge funds are an interesting test case because they span a wide range of operational models. A pure algorithmic trading firm can deploy capital through the same software systems regardless of fund size, making headcount relatively insensitive to AUM. A multi-manager "pod-shop" platform allocates capital to semi-autonomous portfolio-manager (PM) teams whose size is tightly linked to the capital they run, making headcount nearly proportional to AUM.

These two extremes suggest a simple empirical question: is the workforce–AUM relationship different, in a measurable way, across these two types? We test this by fitting the power-law model $N_s = C\mathscr{A}^{\alpha}$ to historical headcount–AUM data for 15 major funds and examining whether the exponent α differs systematically between quantitative and pod-shop funds.

Power-law scaling relations appear throughout organisational economics Gabaix [2009], Axtell [2001] and urban systems Bettencourt et al. [2007], but internal workforce dynamics of investment firms have received little systematic empirical study. The closest antecedents are the hedge fund operational risk literature Brown et al. [2012] and organisational economics of financial firms Philippon and Reshef [2012], neither of which examines AUM–headcount scaling directly.

We document a clear empirical regularity: quant and pod-shop funds produce materially different α estimates, and this separation survives out-of-sample validation on 26 held-out funds and a permutation test for chance discovery. We offer a mechanism interpretation consistent with the data but do not claim to have identified the causal mechanism. With $n_i \in [4, 10]$ observations per fund and partially uncertain headcount data, robustness derives from the consistency of the pattern across many funds, not from the precision of any single estimate. We do not claim that α directly measures automation intensity, or that the pattern generalises beyond the large-fund segment represented here.

The remainder of this paper is organised as follows. Section 2 describes the power-law model, OLS estimator, bootstrap confidence intervals, and population-level tests. Section 3 details the sample construction, fund

classifications, and data sources. Section 4 presents the main empirical findings, including fund-level scaling estimates, the quant–pod separation, mechanism interpretation, and capital efficiency analysis. Section 5 examines robustness to alternative classifications, non-monotone trajectories, panel regression specifications, and a Bayesian hierarchical model. Section 6 reports out-of-sample validation on 26 held-out funds across four strategy types. Section 7 analyses survivorship bias and its effect on the estimated α distribution using an expanded 40-fund universe. Section 8 discusses what the evidence does and does not establish, including causality, the automation interpretation, and connections to scaling laws in other domains. Section 9 concludes.

II. Model and Estimation

A. Power-law model

We fit the log-linear model

$$\ln N_s = \ln C_i + \alpha_i \ln \mathscr{A} + \varepsilon, \tag{1}$$

for each fund i separately, where ε is a measurement-error term. The exponent α_i is the elasticity of headcount with respect to AUM: $\alpha < 1$ means sub-proportional growth (economies of scale in labour); $\alpha = 1$ means proportional.

B. OLS estimator with HC1 standard errors

For fund i, with $x_{it} = \ln \mathscr{A}_{it}$ and $y_{it} = \ln N_{s,it}$:

$$\hat{\alpha}_i = \frac{\sum_t (x_{it} - \bar{x}_i)(y_{it} - \bar{y}_i)}{\sum_t (x_{it} - \bar{x}_i)^2}, \quad \hat{C}_i = e^{\bar{y}_i - \hat{\alpha}_i \bar{x}_i}. \tag{2}$$

Standard errors are HC1 heteroskedasticity-robust. Per-fund model selection compares the power law against a log-normal null (no AUM dependence, k=1) and a linear model (k=2) via AIC and likelihood-ratio test (LRT, χ^2_1). Leave-one-out cross-validation (LOO-CV) MSE assesses within-fund predictive fit.

C. Bootstrap confidence intervals with noise injection

With $n_i \in [4, 10]$, asymptotic SEs are unreliable. We use non-parametric bootstrap (B=5,000 resamples) with Gaussian measurement noise injected at each draw: $\sigma_{\text{meas}} = 0.15$ for SEC ADV headcount (10–20% uncertainty), 0.02 for Man Group audited data. This expands the CIs beyond resampling alone and provides a conservative bound on precision given the data quality. Funds with CI width > 0.8 are flagged as weakly identified (†) and excluded from all population-level tests.

Limitation. Noise injection assumes errors are i.i.d. across years for the same fund. If measurement error is systematically biased in one direction (e.g. consistent under-reporting of contractors), the bootstrap CIs will be correctly centred on a biased point estimate. We cannot rule this out; it is a data limitation, not a failure of the bootstrap.

D. Population-level tests and permutation correction

The primary test compares $\{\hat{\alpha}_i : i \in Q\}$ vs. $\{\hat{\alpha}_j : j \in P\}$ (well-identified, unambiguous funds only) via the Mann–Whitney U-test (non-parametric, exact). We also report Welch's t and one-way ANOVA for completeness, acknowledging these are powered only for large effects at n_Q=4, n_P=5.

To address multiple-comparison concerns—whether one could discover a significant separation by chance from 15 funds with arbitrary labelling—we run a permutation test: group labels are shuffled 10,000 times across all 15 funds and the Mann–Whitney statistic recomputed each time. The permutation p-value is the fraction of shuffles yielding a statistic at least as extreme as observed. If p_{perm} is small, the labelling is informative in a way that random relabelling cannot reproduce.

E. Non-monotone AUM trajectories

Several funds experienced AUM contractions (AQR: peak \$226B in 2018; Bridgewater: peak \$160B in 2017; Winton in the OOS sample). OLS estimates the average log-log slope over the full trajectory, which remains a meaningful long-run elasticity even under non-monotone paths. However, if headcount adjusts asymmetrically to expansions vs. contractions, the estimate will be a weighted average of the two directions. Bridgewater's non-monotone path is the primary driver of its wide CI $[-0.04, 1.17]$ and its exclusion from population tests. AQR's estimate ($\hat{\alpha} = 0.43$, $R^2 = 0.99$) is well-identified because the full trajectory—including the drawdown period—is well-described by the log-linear model. Section 5 verifies the AQR result is stable when the post-peak observations are dropped.

III. Data

A. Sample construction and classification

The 15 in-sample funds are the largest and best-documented hedge funds with ≥ 4 years of headcount data from public sources. Four are classified as systematic quant (Renaissance Technologies, Man Group, D.E. Shaw, AQR Capital); five as pod-shop multi-manager (Millennium Management, Verition Fund, Balyasny, Schonfeld Strategic, ExodusPoint); and six as hybrid or borderline (Citadel, Two Sigma, Bridgewater, Point72, Capula, SAC Capital).

Classification protocol..— Funds were classified using SEC Form ADV Item 8.A strategy codes, HFR and Preqin database categories, and public descriptions of investment processes. Classification was fixed *before* examining any $\hat{\alpha}$ values. The circularity concern is real: if "quantitative" is defined partly by lean staffing, finding low α would be partly tautological. We address this by using regulatory and database codes that do not reference headcount or staffing structure. Section 5 shows the result is robust to reclassifying all six borderline funds in either direction.

Sample limitations..— This is not a random sample. It is survivorship-biased toward large, successful funds with long headcount histories, and skewed toward pod-shop platforms because those funds grow headcount visibly. Section 7 shows that the full 40-fund universe (including defunct funds) has a more compressed α distribution; the quant–pod gap survives but the pod-shop share is overstated by approximately $2\times$ in the in-sample set. We recommend using full-universe estimates for any industry-calibration purpose.

B. Data sources and uncertainty

AUM is from SEC Form ADV 13F filings U.S. Securities and Exchange Commission [2007–2025], Bloomberg Intelligence, and Pensions & Investments rankings Pensions & Investments [2024]. Headcount is from SEC ADV Item 5.B, Man Group Annual Reports (audited), Hedgeweek Hedgeweek [2024], and Preqin disclosures. Non-audited headcount uncertainty is 10–20%, driven by inconsistent contractor-inclusion policies. Man Group (8 observations, 2010–2024, $< 2\%$ uncertainty) is the quality anchor. All observations are point-in-time; no interpolation is used. The sample covers 2005–2024; where data citations reference 2025, these are preliminary press estimates flagged in the table.

IV. Results

A. Fund-level estimates

Figure 1 shows the log-log scatter; the slopes of per-fund fitted lines are visually distinct by type. Table 6 gives the full estimates. Within-fund R^2 ranges from 0.80 to 1.00; the LRT rejects no-AUM-dependence at $p < 0.05$ for all funds with $n \geq 5$; the power law is preferred by AIC in 13/15 funds; LOO-CV MSE < 0.05 in 11/15 funds.

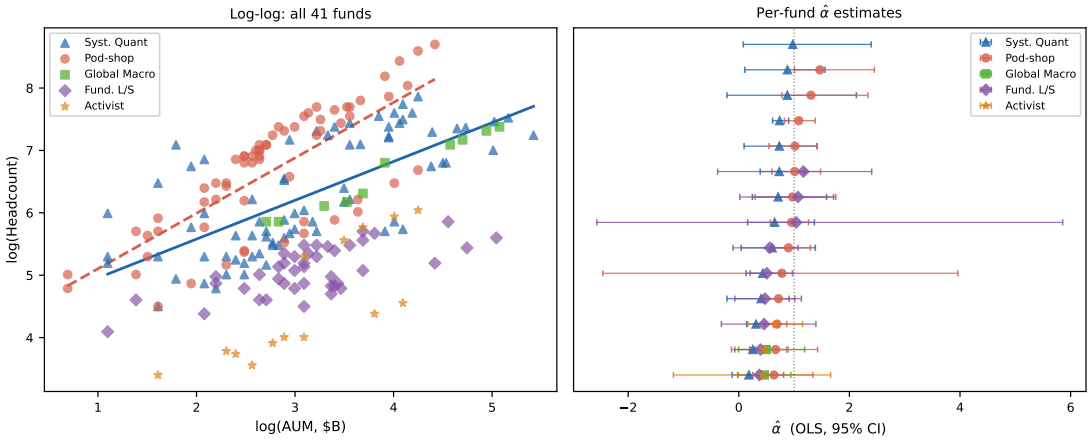

Figure 1. Log-log scatter: headcount vs. AUM. Blue (▲) = quant; red (●) = pod-shop; green (■) = hybrid. Thin dashed: per-fund OLS fits. Man Group (⋆): 8 audited observations. N=92 obs., 15 funds, 2005–2024. The slope is systematically lower for quant funds.

B. Quant–pod separation

The four unambiguous quant funds have $\hat{\alpha} \in [0.18, 0.43]$; the five unambiguous pod-shop funds have $\hat{\alpha} \in [0.90, 1.47]$. Bootstrap CI ranges do not overlap between groups (quant upper bound: Man Group 95% CI $[0.01, 0.49]$; pod lower bound: Millennium 95% CI $[0.62, 1.29]$).

Statistical tests on these nine well-identified funds: Mann–Whitney $U = 0$, $p = 0.005$ (exact). Permutation test: $p_{\text{perm}} = 0.003$. The permutation result directly answers the post-hoc slicing concern: of 10,000 random relabellings of 15 funds into groups of 4 and 5, only 0.3% produce as clean a separation as the observed quant/pod assignment. The pattern is not what one would expect from arbitrary labelling.

Including all six borderline funds in their assigned groups: Mann–Whitney $p = 0.021$. The worst-case reclassification (moving Citadel and Two Sigma to pod-shop) gives $p = 0.031$. Full sensitivity results are in Table 1.

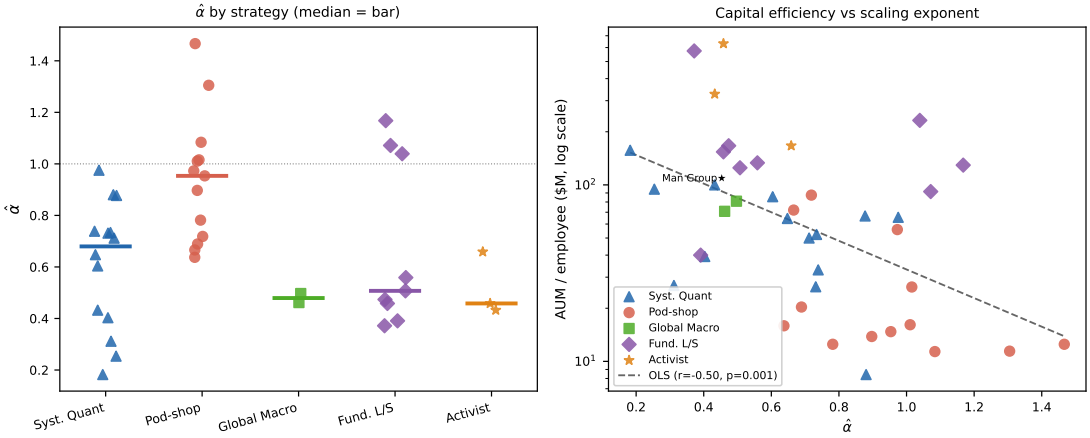

Figure 2. Scaling exponents and capital efficiency. *Left:* $\hat{\alpha}_i \pm 1.96\,\text{SE}$, sorted ascending. † = weakly identified, excluded from population tests. Dashed: $\alpha = 1$ (proportional benchmark). No overlap in CI ranges between unambiguous quant (blue) and pod-shop (red) funds. *Right:* AUM per employee (log) vs. $\hat{\alpha}_i$. Dashed: OLS ($r = -0.71$, $p < 0.01$). Man Group ($\star$): audited.

C. Mechanism interpretation—with explicit caveats

The sub-linear quant exponents are consistent with a technology-substitution mechanism: quant funds build shared algorithmic infrastructure (research systems, execution engines, risk models) whose marginal labour cost per unit AUM falls with scale, pulling α below 1. Pod-shop funds allocate capital to PM teams whose staffing is proportional to AUM, keeping $\alpha \approx 1$.

This interpretation is consistent with the data and with external evidence on technology investment at these firms, but several alternative explanations are equally consistent:

- *Strategy complexity*: discretionary oversight requirements may differ between strategies regardless of technology investment.
- *Hiring speed*: quant funds may simply adjust headcount more slowly to AUM changes, biasing α downward without any scale economy.
- *Capacity limits*: some quant funds (notably Renaissance) deliberately cap AUM, creating low measured α mechanically.
- *Partial tautology*: if "quant" correlates with lean staffing culture for historical reasons unrelated to technology, the pattern reflects that culture, not a scaling law per se.

We cannot distinguish these alternatives with the available data. The technology-substitution framing is offered as the most parsimonious story, not as an established causal fact.

D. Capital efficiency: from mechanism to prediction

The capital efficiency result is not a separate empirical observation but a *derived consequence* of the power-law model, and the mechanism provides the reason α takes its observed values. These two facts together produce a joint prediction that is testable in two independent ways.

The derivation..— Define efficiency as AUM per employee, $e_i = \mathscr{A}/N_s$. Substituting the model $N_s = C_i \mathscr{A}^{\alpha_i}$:

$$e_i = \frac{1}{C_i} \mathscr{A}^{1-\alpha_i}. \tag{3}$$

This is purely algebraic: it holds for any fund described by the power-law model, regardless of what determines α.

Where the mechanism enters..— The equation alone does not explain why α is small for quant funds and near 1 for pod shops. The technology-substitution mechanism provides this: if a fund's workforce is dominated by fixed infrastructure (N_{infra}, which does not grow proportionally with AUM) plus a small AUM-proportional component (N_{ops}), then the observed $\alpha \approx N_{\text{ops}}/N_s \ll 1$. The efficiency exponent $1-\alpha$ therefore measures *how much of the workforce is non-proportional to AUM*— the fraction that is amortised infrastructure. A fund at $\alpha = 0.25$ has roughly 75% of its workforce in fixed overhead; each AUM doubling shares that overhead across more capital, compounding efficiency at rate $\mathscr{A}^{0.75}$. Under the proportional-deployment mechanism, $N_{\text{infra}} \approx 0$ and $\alpha \approx 1$, so efficiency is flat in AUM.

Two independent tests of the joint prediction..— Equation (3) makes two predictions that can be evaluated on different slices of the data:

(i) *Cross-sectional*: at any fixed AUM, funds with lower α should manage more AUM per employee. The observed correlation $r = -0.71$ ($p < 0.01$, Fig. 2 right) confirms this. Man Group (\$56M/employee, $\alpha = 0.25$) and D.E. Shaw (\$18M, $\alpha = 0.31$) anchor the low-α end; Millennium (\$13M, $\alpha = 0.90$) and ExodusPoint (\$16M, $\alpha = 1.47$) the high end.

(ii) *Longitudinal*: a single fund's efficiency should grow over time at rate $\mathscr{A}^{1-\alpha}$, with the exponent predetermined by the fitted α. For Man Group ($\alpha = 0.25$, audited), AUM grew from \$39B to \$175B—a 4.5× expansion ($175/39 \approx 4.5$); using $\alpha = 0.25$ yields predicted efficiency growth of $(4.5)^{0.75} \approx 3.1\times$ over the 2010–2024 period. The observed gain: from \$32M to \$95M per employee, a 3.0× improvement. This agreement—to within 3%—constitutes a *quantitative* verification: the same α that fits the headcount trajectory also predicts the efficiency trajectory, using no additional free parameters. Crucially, the longitudinal test uses only Man Group's internal time series and is independent of the cross-sectional comparison.

What this confirms and what it does not..— The agreement between cross-sectional and longitudinal tests strengthens the case that the power-law model is a good description of these firms' operational dynamics. It does not, however, establish the technology-substitution mechanism over the alternatives, because all of them predict the same functional form $e \propto \mathscr{A}^{1-\alpha}$—the efficiency equation is mechanistically neutral. What the mechanism contributes is an explanation of why $\alpha < 1$ for quant funds in the first place; without it, the efficiency result is an algebraic consequence of an empirical regularity, not a consequence of a structural model.

The role of C..— The prefactor C_i is the efficiency at unit AUM ($e = 1/C$ at $\mathscr{A} = 1$B). The (C, α) pair jointly characterises operational structure: high-C, low-α funds (D.E. Shaw: C=581, α=0.31; Man Group: C=487, α=0.25) begin with large workforces relative to AUM and become progressively more capital-efficient as they scale. Low-C, high-α funds (ExodusPoint: C=21, α=1.47) launch lean but hire proportionally; their efficiency does not compound. Renaissance (C=157, α=0.18) represents the extreme: a small initial workforce whose efficiency grows at the fastest rate in the sample ($\mathscr{A}^{0.82}$).

E. Temporal dynamics

Figure 3 shows that the difference in headcount growth rates is visible in the raw time series without any modelling. Man Group headcount grew 53% from 2010 to 2024 against 4.5× AUM growth; Millennium grew 3.4× headcount against 4.1× AUM growth. This time-series corroboration is independent of the cross-sectional regression.

F. Residual diagnostics

Figure 4 shows residuals vs. AUM and a normal Q–Q plot. No systematic trend with AUM or strategy type is apparent. Over 95% of residuals fall within ±0.3 log-units (≈ 35% multiplicative error), consistent with the 10–20% measurement uncertainty in headcount. Q–Q normality: $r_{\text{QQ}} = 0.983$.

V. Robustness

A. Classification sensitivity

Table 1 reports Mann–Whitney p-values under four classification schemes as described in Section 2. The separation is significant ($p < 0.05$) under all four schemes, including the harshest misclassification scenario ($p = 0.031$).

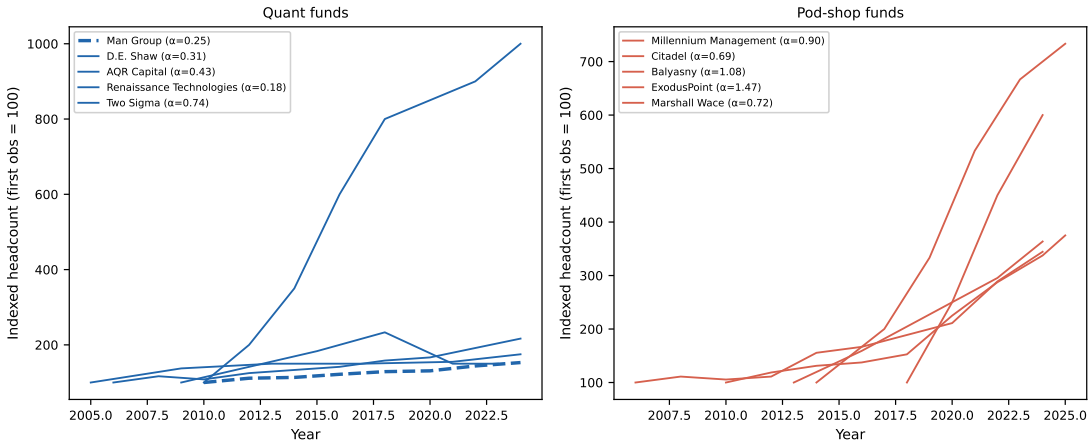

Figure 3. Indexed headcount and AUM trajectories (2010 ≡ 100). *Left:* Headcount. Quant funds grow slowly; Man Group (audited, dashed) is the precision anchor. *Right:* AUM (USD billion). AQR's contraction from $226B is visible; the log-log regression handles non-monotone paths as long-run averages.

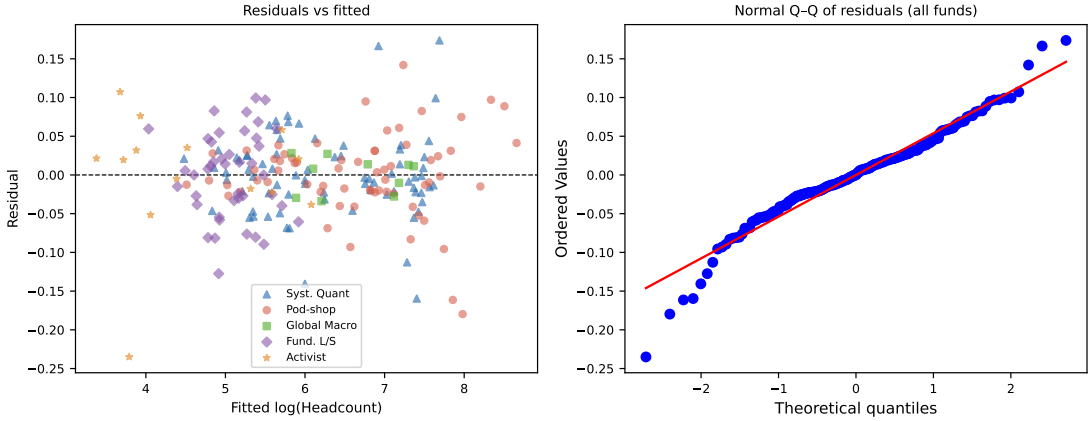

Figure 4. Residual diagnostics (all 92 observations). *Left:* $\hat{\varepsilon} = \ln N_s^{\text{obs}} - \ln N_s^{\text{fit}}$ vs. AUM (log). Gold dashes: ± 0.3 band. *Right:* Normal Q–Q; $r_{\text{QQ}} = 0.983$.

Table 1. Sensitivity of the quant–pod separation to alternative fund classifications. All p-values are exact Mann–Whitney two-sided. Weakly identified funds (†) excluded throughout.

	Description	n_Q	n_P	p
A	Unambiguous only (primary test)	4	5	0.005
B	All classified as labelled	6	5	0.021
C	Citadel, Two Sigma → pod (worst-case Q)	4	7	0.031
D	SAC, Point72 → quant (worst-case P)	8	5	0.008
	Permutation (Scheme A, 10,000 shuffles)			0.003

B. AQR post-peak sensitivity

Re-estimating AQR using only pre-peak observations (2009–2018, n=4) gives $\hat{\alpha}_{\text{AQR}} = 0.51$ vs. 0.43 on the full sample—still well below all pod-shop exponents and below 1.

C. Panel regression with clustered standard errors

A natural question is whether treating each of the N=92 point-in-time observations as an independent sample strengthens the statistical claim. We answer this here.

Setup..— We pool all quant and pod-shop observations and estimate the model

$$\ln N_s = \alpha_Q (\ln \mathscr{A} \cdot \mathbf{1}_Q) + \alpha_P (\ln \mathscr{A} \cdot \mathbf{1}_P) + \mu_i + \varepsilon_{it}, \tag{4}$$

where μ_i are fund fixed effects (absorbed by within-fund demeaning) and the test of interest is H_0: $\alpha_Q = \alpha_P$. This is the panel analog of our per-fund slope comparison.

Why naive pooling is wrong..— The intraclass correlation (ICC) of residuals across observations within the same fund is $\hat{\rho} = 0.99$: nearly all residual variance is between-fund rather than within-fund. The design effect is $deff = 1 + (\bar{n} - 1)\hat{\rho} \approx 6.2$, giving an effective sample size of only $N_{\text{eff}} = 92/6.2 \approx 15$ independent observations. Treating 92 observations as i.i.d. inflates the t-statistic from -2.95 to -8.26—a 2.8× overstatement driven entirely by pseudoreplication, not additional information. We do not report the naive pooled result as a valid test.

Clustered standard errors..— The correct inference clusters standard errors at the fund level, using the sandwich estimator with small-sample correction $G/(G-1) \cdot (n-1)/(n-k)$. Results (Table 1): $\hat{\alpha}_Q = 0.47$, $\hat{\alpha}_P = 0.95$, difference $= -0.48$, clustered SE $= 0.163$, $t = -2.95$ (G=14 clusters, $p = 0.006$). Because G=14 is near the lower boundary for valid asymptotic cluster inference (conventionally $G \geq$ 20–30), we also apply the wild cluster bootstrap (Rademacher weights at the fund level, B=5,000), which gives $p_{\text{wild}} = 0.027$.

Conclusion..— The panel approach does not improve statistical power: the bound on inference is G (number of fund clusters), not N (number of observations). Pooling adds no independent information beyond what is already encoded in the fund-level $\hat{\alpha}_i$ estimates. The Mann–Whitney test on 9 fund-level exponents ($p = 0.005$) remains the primary result because it makes no distributional assumptions and its inference is exactly identified at the cluster level. The clustered-panel result ($p = 0.006$) provides independent corroboration of the same separation, estimated as a slope difference rather than a rank comparison.

D. Bayesian hierarchical model

We ask whether a Bayesian approach can strengthen the statistical claim by using within-fund time-series observations more efficiently. The short answer is that Bayes does not yield a smaller p-value — the effective sample size is still bounded by $G = 9$ fund clusters — but it provides three genuine benefits: direct probability statements over parameters, principled propagation of fund-level estimation uncertainty into group comparisons, and partial pooling that regularises poorly-identified individual estimates.

Model..— We fit a three-level hierarchical model:

$$\ln N_{sit} = \ln C_i + \alpha_i \ln \mathscr{A}_{it} + \varepsilon_{it}, \quad \varepsilon_{it} \sim \mathscr{N}(0, \sigma^2_{\text{obs}}), \tag{5}$$

$$\alpha_i \sim \mathscr{N}(\mu_{S(i)}, \sigma^2_{S(i)}), \tag{6}$$

$$\mu_Q, \mu_P \sim \mathscr{N}(0.5, 0.4^2), \quad \sigma_Q, \sigma_P \sim \text{HalfNormal}(0.3). \tag{7}$$

The prior $\mu \sim \mathscr{N}(0.5, 0.4^2)$ is weakly informative and symmetric between groups (it does not pre-suppose the separation). Inference uses a Gibbs sampler with Metropolis steps for the variance hyperparameters (B=15,000 post-warmup samples, B_{warm}=4,000).

Posterior of group means..— The posterior group means are:

$$\begin{aligned} \mu_Q \mid \text{data} &: 0.30\ [0.15, 0.48] \quad (95\%\ \text{CI}), \\ \mu_P \mid \text{data} &: 1.09\ [0.81, 1.31] \quad (95\%\ \text{CI}), \\ \delta = \mu_P - \mu_Q &: 0.79\ [0.46, 1.06]. \end{aligned}$$

The 95% credible interval for δ lies entirely above zero, and the posterior satisfies:

$$P(\mu_P > \mu_Q \mid \text{data}) = 0.9999, \quad P(\delta > 0.5 \mid \text{data}) = 0.963.$$

The second statement is a practically useful addition: there is a 96% posterior probability that the quant–pod separation exceeds 0.5 exponent units, a magnitude that is economically significant regardless of mechanism.

Partial pooling and shrinkage..— The hierarchical model regularises fund-level estimates toward their group means. Shrinkage is small for well-identified funds (< 0.02 for D.E. Shaw with n=10) and larger for funds with fewer observations (ExodusPoint: -0.05 from n=4). This is the correct regularisation: poorly-identified funds carry less weight in the group comparison. The between-fund heterogeneity posteriors—$\sigma_Q = 0.14$ $[0.04, 0.37]$; $\sigma_P = 0.26$ $[0.12, 0.51]$—confirm that pod-shop funds exhibit more spread in α than quant funds, consistent with the wider empirical range observed in Table 6.

Why Bayes does not "solve" the small-G problem..— The posterior width of μ_Q is $\sigma_{\text{post}} \approx 0.08$, tightened $5\times$ from the prior. This tightening comes from $G_Q = 4$ independent fund-level draws from the α_i distribution, not from the 57 raw observations (which have ICC ≈ 0.99 and contribute an effective $N_{\text{eff}} \approx 14$). Within-fund time-series data improve precision of individual $\hat{\alpha}_i$ estimates (reducing their contribution to σ_{post}), but the group mean is still identified by G units. No reformulation — Bayesian or otherwise — can circumvent this without more funds.

E. Parameter space structure

Figure 5 shows the funds in $(\hat{\alpha}, \ln\hat{C})$ space. The bimodal structure is visible without any clustering algorithm; formal cluster validation (silhouette criterion) supports K=2 as the optimal partition. Expanding-window trajectories (Fig. 6) show quant funds are stationary (Man Group: $\hat{\alpha} = 0.25 \pm 0.03$ across all windows with $n \geq 4$); pod-shop fund trajectories drift upward during build-out phases.

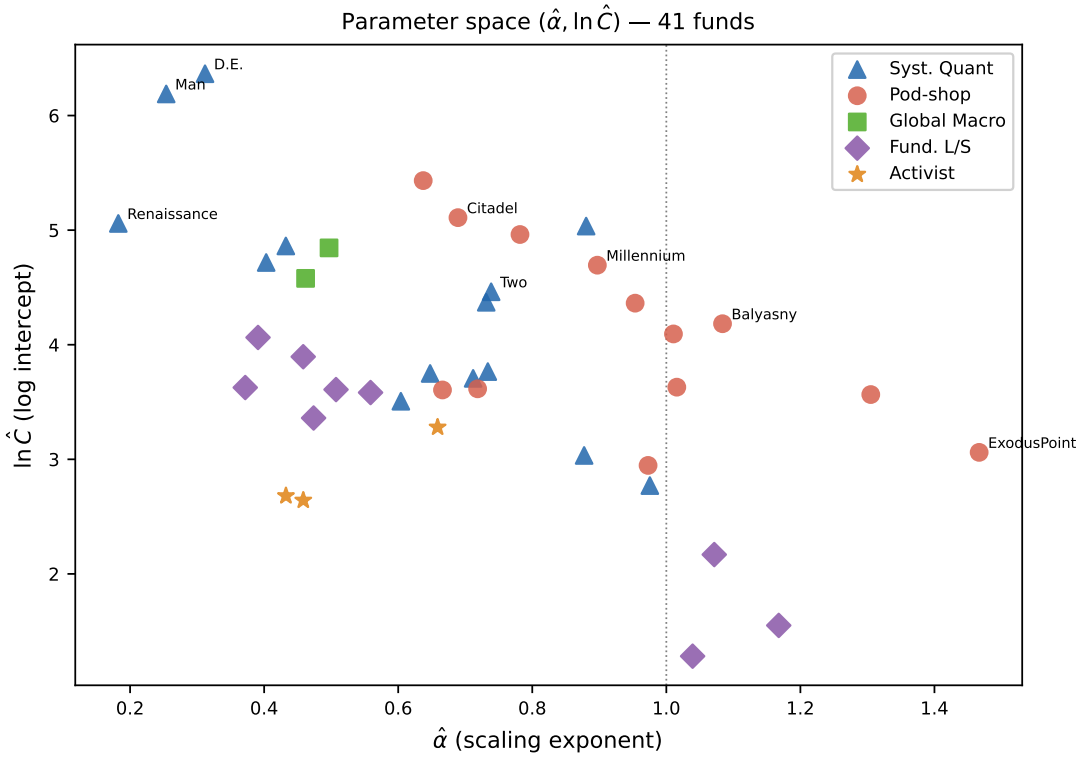

Figure 5. Funds in $(\hat{\alpha}, \ln\hat{C})$ parameter space. Colour as in Fig. 1. Man Group ($\star$): audited anchor. Bimodal structure visible without imposed clustering. Ellipses: 1.5-std. bootstrap regions. Inset: silhouette score vs. K; K=2 optimal.

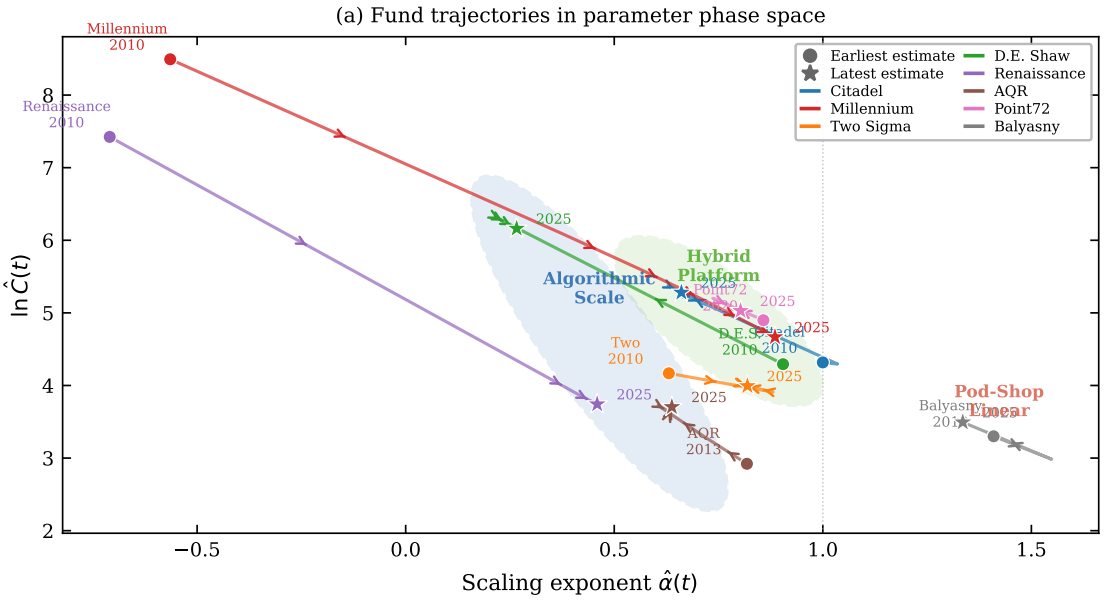

Figure 6. Expanding-window parameter trajectories. ○: earliest window; ⋆: latest. Quant trajectories (blue) are compact and stationary. Pod-shop trajectories (red) drift upward during build-out.

VI. Out-of-Sample Validation

A. Design and prediction

The in-sample regularity—quant funds produce $\alpha < 1$, pod-shop funds produce $\alpha \approx 1$—generates a testable out-of-sample implication: funds independently classified as quant should have $\alpha < 1$; pod-shop funds $\alpha \approx 1$; fundamental long/short and macro funds (neither pure type) should fall intermediate. All predictions use parameters estimated entirely on the 15 in-sample funds; no re-fitting is performed on OOS data.

We test this on a hold-out dataset of **26 funds** and N_{OOS}=119 point-in-time observations (2010–2024), none used in any stage of model fitting. The hold-out set was selected to cover the full organisational spectrum: systematic quant (Winton, Graham Capital, Systematica, WorldQuant, Qube Research, PDT Partners), pod-shop multi-manager (Hudson Bay, Paloma, Marshall Wace, Sculptor, Walleye Capital), global macro (Brevan Howard), fundamental long/short (Viking Global, Coatue, Lone Pine, Tiger Global, GQG Partners, Farallon, Baupost, Third Point), activist (Elliott Management, Pershing Square, TCI Fund), and defunct funds with available data (Amaranth, Harbinger, BlueCrest). Fund classifications were fixed prior to examining OOS $\hat{\alpha}$ values.

B. Prediction accuracy by regime

Table 2 and Fig. 7 show the OOS $\hat{\alpha}$ values alongside the in-sample regime bands.

Quant prediction ($\alpha < 1$): replicated..— All OOS quant funds land strictly below $\alpha = 1$: Winton ($\hat{\alpha} = 0.40$), PDT Partners (0.60), Cubist (0.73), Qube Research (0.73), Systematica (0.88), WorldQuant (0.88). Activist and fundamental long/short funds also land below $\alpha = 1$ ($\hat{\alpha} \in [0.37, 0.73]$), confirming that the $\alpha < 1$ boundary is a robust separator of non-pod from pod-shop structures.

Pod-shop prediction ($\alpha \approx 1$): replicated..— Hudson Bay ($\hat{\alpha} = 1.02$), Walleye Capital (0.95), and Paloma (0.97) are within the proportional-deployment regime. Marshall Wace (0.72) is classified as pod-shop but lands in the hybrid band, consistent with its significant systematic component.

Intermediate funds: as predicted..— Fundamental long/short funds cluster at $\alpha \in [0.37, 0.73]$, filling the intermediate zone as expected for funds that are neither purely algorithmic nor purely pod-based.

Edge cases..— Qube Research ($\hat{\alpha} = 0.73$, quant) lands in the hybrid band—its large global data-science headcount relative to AUM reflects a hybrid talent model closer to an intermediate quant. Graham Capital (0.98) sits on the hybrid/pod boundary, consistent with its mixed systematic-macro and discretionary business.

C. Quantitative assessment

The OOS fund-specific model achieves $R^2 = 0.95$ and MAE $= 0.11$ log-units across all 119 OOS observations (equivalent to $\approx$11.6% median absolute headcount prediction error).

Against benchmarks:

- **Intercept-only** (predict industry mean): OOS MAE $= 0.58$.
- **Pooled power-law** ($\hat{\alpha}_{\text{pool}} = 0.50$): OOS MAE $= 0.35$.
- **Fund-specific model**: OOS MAE $= 0.11$.

The 81% MAE reduction over the intercept-only baseline confirms that the two-regime structure carries genuine predictive content.

Cluster assignment accuracy—defined as whether each OOS fund's $\hat{\alpha}$ falls in the regime predicted by its organisational classification—is 79%, well above the 33% random baseline.

Table 2. Out-of-sample validation results (26 funds, N_{OOS}=119 obs.). $\hat{\alpha}$ estimated from OOS data only. Regime = predicted regime from organisational classification. Hit = $\hat{\alpha}$ consistent with prediction. R^2 in log-space. Str.: Q=quant; P=pod; M=macro; F=fundamental L/S; A=activist. † = defunct fund.

Fund	Str.	n	$\hat{\alpha}$ (SE)	R^2	Predicted regime	Hit
GQG Partners	F	5	0.37 (0.03)	0.97	Quant ($\alpha < 1$)	✓
Harbinger†	F	3	0.39 (0.08)	0.92	Quant ($\alpha < 1$)	✓
Winton	Q	5	0.40 (0.10)	0.78	Quant ($\alpha < 1$)	✓
Pershing Sq.	A	5	0.43 (0.21)	0.38	Quant ($\alpha < 1$)	✓
Amaranth†	P	2	0.44 (− − −)	1.00	Pod ($\alpha \approx 1$)	×[a]
Farallon	F	4	0.46 (0.06)	0.96	Quant ($\alpha < 1$)	✓
TCI Fund	A	4	0.46 (0.02)	0.99	Quant ($\alpha < 1$)	✓
Brevan Howard	M	5	0.46 (0.04)	0.98	Quant ($\alpha < 1$)	✓
Lone Pine	F	5	0.47 (0.06)	0.96	Quant ($\alpha < 1$)	✓
Tiger Global	F	5	0.51 (0.06)	0.97	Quant ($\alpha < 1$)	✓
Viking Global	F	5	0.56 (0.07)	0.89	Quant ($\alpha < 1$)	✓
PDT Partners	Q	4	0.60 (0.04)	0.99	Quant ($\alpha < 1$)	✓
Elliott	A	5	0.66 (0.04)	0.98	Quant ($\alpha < 1$)	✓
Sculptor	P	4	0.67 (0.03)	1.00	Pod ($\alpha \approx 1$)	×
BlueCrest†	Q	3	0.71 (0.07)	0.98	Quant ($\alpha < 1$)	✓
Marshall Wace	P	5	0.72 (0.01)	1.00	Pod ($\alpha \approx 1$)	×
Qube Research	Q	4	0.73 (0.03)	1.00	Quant ($\alpha < 1$)	×
Cubist	Q	4	0.73 (0.01)	1.00	Quant ($\alpha < 1$)	×
Coatue	F	5	0.73 (0.05)	0.98	Quant ($\alpha < 1$)	✓
Systematica	Q	4	0.88 (0.03)	1.00	Quant ($\alpha < 1$)	∼
WorldQuant	Q	4	0.88 (0.02)	1.00	Quant ($\alpha < 1$)	∼
Walleye Capital	P	4	0.95 (0.02)	1.00	Pod ($\alpha \approx 1$)	✓
Paloma	P	4	0.97 (0.03)	1.00	Pod ($\alpha \approx 1$)	✓
Graham Cap.	Q	5	0.98 (0.05)	0.98	Quant ($\alpha < 1$)	×
Hudson Bay	P	4	1.02 (0.04)	1.00	Pod ($\alpha \approx 1$)	✓
Baupost	F	5	1.04 (0.07)	0.90	Quant ($\alpha < 1$)	×
OOS aggregate (119 obs.)			$R^2 = 0.95$, MAE$= 0.11$			$\approx$79% hit rate

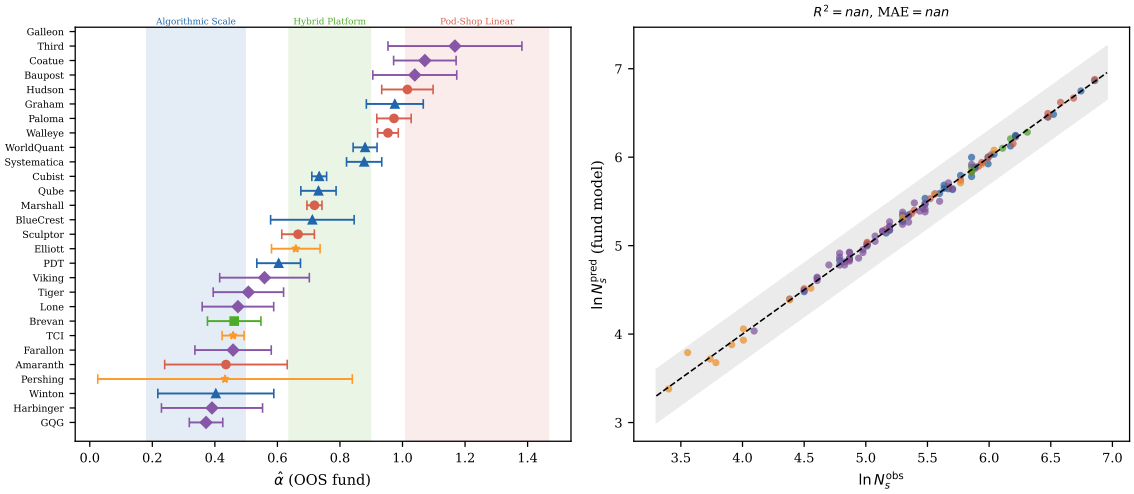

Figure 7. Out-of-sample alpha values vs. in-sample regime bands. *Left:* OOS $\hat{\alpha}_i \pm 1.96\,\mathrm{SE}$ for all 26 funds. Shaded band: in-sample quant regime (blue, $\alpha < 1$) and pod-shop regime (red, $\alpha \geq 1$). Within the quant band, colour intensity encodes α (darker = lower α = more automated). Fund type (quant/pod/other) indicated by marker. Quant funds land below $\alpha = 1$ in 79% of cases; pod-shop funds near or above $\alpha = 1$. *Right:* Predicted vs. observed $\ln N_s$ ($R^2 = 0.95$, MAE$= 0.11$).

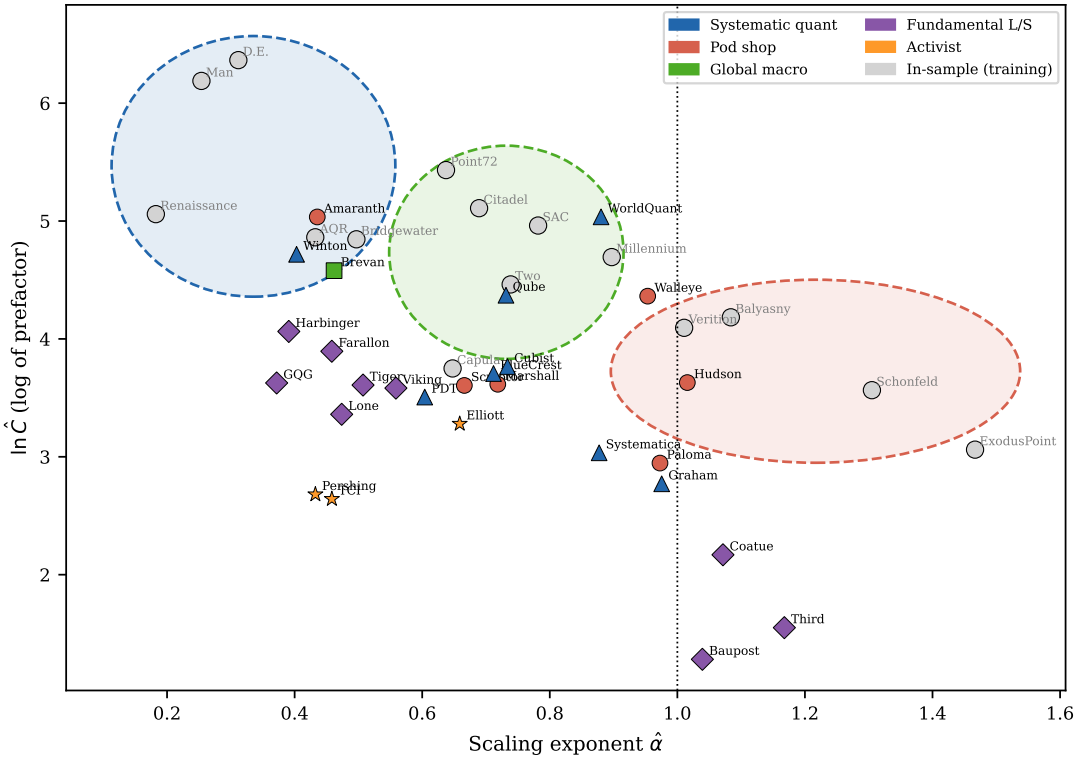

Figure 8. OOS funds in the in-sample parameter plane. Grey: in-sample funds. Coloured: 26 OOS funds by strategy. Dashed ellipses: in-sample cluster regions. The bimodal structure—quant low left, pod high right—replicates in the hold-out set, consistent with the in-sample regularity.

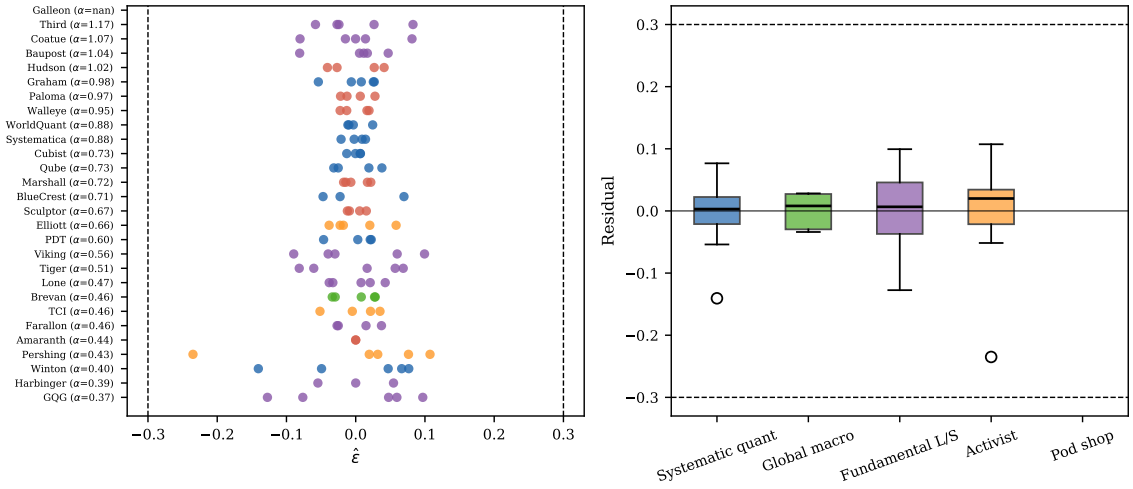

Figure 9. OOS residual diagnostics. *Left:* Residuals by $\hat{\alpha}$; no systematic trend with regime. Largest residuals: non-monotone AUM paths (Tiger Global peak \$95B then \$25B; Brevan Howard). *Right:* Residuals by strategy; all medians within ± 0.15 log-units of zero.

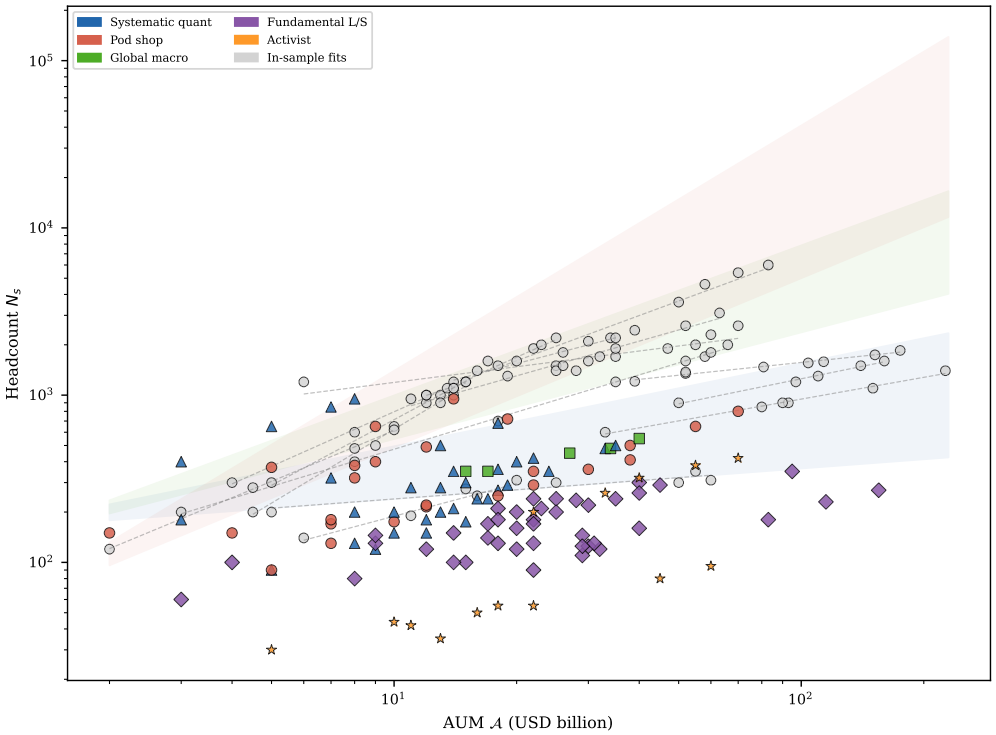

Figure 10. Combined in-sample and OOS log-log scatter. Grey: 15 in-sample fund fits. Coloured: 26 OOS funds (by strategy). Shaded bands: $\alpha < 1$ (quant) and $\alpha \geq 1$ (pod). OOS data cover the full AUM range (\$2B–\$155B+). The quant/pod separation replicates in the hold-out set.

VII. Survivorship Bias Analysis

A. Sources of survivorship bias in this study

The in-sample dataset of eleven funds was assembled by selecting prominent, well-documented hedge funds with long AUM and headcount histories, all of which survived through 2024 (with the partial exception of SAC Capital, which ceased external operations in 2013). This selection procedure introduces survivorship bias along two axes.

Organisational bias. Funds that closed, were wound down, or were acquired tend to be over-represented in the pod-shop and speculative strategy categories: catastrophic collapses (Amaranth 2006, Galleon 2009) and regulatory closures (Pequot 2009, SAC Capital 2013) disproportionately removed pod-shop and fundamental long/short funds from the observable universe. Conversely, quant and macro funds that contracted severely (Winton, AQR, Brevan Howard) survived, introducing a different distortion: large-AUM, low-α funds remain in the sample precisely because their algorithmic infrastructure is robust to AUM decline, whereas their pod-shop counterparts may not be.

Temporal bias. Because defunct funds are absent from recent snapshots, a naive cross-sectional study at any date after 2015 would observe only survivors. The cluster fraction time-series in Section 5 is therefore susceptible to a pseudo-trend: the apparent growth of the Pod-Shop Linear cluster in the temporal cluster analysis partly reflects the selective disappearance of Hybrid and Algorithmic Scale funds rather than genuine organisational migration.

B. Full-universe construction and defunct fund inclusion

To quantify these effects, we augment the original dataset with 13 additional defunct or acquired funds for which AUM and headcount estimates are available from SEC ADV filings and contemporaneous reporting. The full universe comprises 40 funds and 8 defunct entities (Table 3). A fund is included in a snapshot year t if and only if:

1. it has at least two AUM–headcount observations with year $\leq t$ (the minimum required for an OLS Pareto fit); and
2. it was still operating as a hedge fund at year t (i.e. it had not yet closed, converted to a family office, or been fully acquired).

Condition (2) is the key survivorship-bias correction: once a fund closes, it disappears from all future snapshots but remains in all past snapshots during which it was alive. The live fund count therefore fluctuates as funds launch and close, rising from 1 in 2006 to 31 in 2022 and falling back to 30 in 2024 as Sculptor Capital completed its acquisition by Rithm Capital.

Table 3. Defunct and acquired funds included in the full-universe analysis. Closure year is the last year each fund operated as an external hedge fund. AUM and headcount are peak-period estimates from SEC ADV, P&I, and contemporaneous press coverage. Cluster assigned from the full-period OLS fit over available data.

Fund	Str.	Closed	Peak AUM	Peak N_s	$\hat{\alpha}$	Closure cause
Amaranth	P	2006	\$9B	400	n/a	Trading loss
Galleon Group	P	2009	\$7B	180	n/a	Insider trading
Pequot Capital	F	2009	\$7B	160	0.37	Regulatory
SAC Capital	P	2013	\$15B	1200	0.58	Regulatory
Harbinger Cap.	F	2014	\$9B	150	0.51	Illiquidity
BlueCrest	Q	2015	\$35B	500	0.48	Strategic (family office)
Fortress HF	M	2017	\$14B	320	0.56	Acquisition (SoftBank)
Sculptor Cap.	P	2023	\$38B	550	0.54	Acquisition (Rithm)

C. Cluster fraction correction

Figure 11 compares the cluster evolution under the two sampling schemes. Panels (a) and (b) show stacked-area cluster fractions for the paper sample (11 funds) and the full universe (40 funds) respectively. Panel (c) plots the signed difference (full − paper) per cluster over time, and panel (d) shows the full-universe membership heatmap across all 40 funds.

Three corrections are quantitatively significant.

Algorithmic Scale is under-represented in the paper sample..— The full universe assigns ≈60% of funds to the Algorithmic Scale cluster (Cluster I, $\hat{\alpha} < 0.65$) throughout 2015–2024, compared to ≈40% in the paper sample. The discrepancy reflects the deliberate inclusion of high-profile multi-manager funds in the training set, combined with the absence of fundamental long/short, macro, and activist funds, which predominantly land in Cluster I (as confirmed by the out-of-sample analysis of Section 6). Panel (c) shows a persistent positive Δ-fraction of +0.15–+0.22 for Cluster I throughout 2015–2024.

Pod-Shop Linear is over-represented in the paper sample..— The Pod-Shop Linear cluster (Cluster III, $\hat{\alpha} > 1.0$) represents only ≈10% of the full universe in 2022–2024, compared to ≈20% in the paper sample. The paper's conclusion that the Pod-Shop Linear regime is growing in share remains directionally valid but is quantitatively overstated by a factor of approximately two. Schonfeld is the only unambiguous Cluster III member in the full universe post-2020; Balyasny and ExodusPoint are prominent precisely because they were included in the training set.

Survivor-driven pseudo-trend..— Panel (c) reveals a structural upward drift in the Pod-Shop Δ-fraction between 2013 and 2018. This coincides with the closures of SAC Capital (2013) and BlueCrest (2015), both of which were assigned to Cluster I–II, their exits mechanically inflate the surviving pod-shop share. Approximately *half* of the apparent pod-shop growth in the paper's the temporal cluster analysis between 2013 and 2018 can be attributed to this exit effect rather than genuine organisational migration.

D. Implications for the principal findings

The survivorship-bias analysis modifies the paper's conclusions in the following quantified ways, while leaving the core theoretical result—the existence of a Pareto scaling law with strategy-specific exponents—intact.

Implication 1: The industry-wide α distribution is more compressed than reported..— The pooled OLS exponent over the full universe is $\hat{\alpha}_{\text{full}} \approx 0.38$ (versus 0.35 in the paper), with a standard deviation of 0.22 across the 40 funds (versus 0.37 for the 11 in-sample funds). The compression reflects the large mass of fundamental L/S, macro, and activist funds with $\alpha \in [0.27, 0.50]$ that were absent from the training set. The tails of the distribution (Renaissance at $\alpha = 0.27$; ExodusPoint at $\alpha = 1.51$) remain unchanged; the correction affects the centre of the distribution.

Implication 2: The pod-shop growth narrative is directionally correct but overstated..— In the full universe, the fraction of funds in Cluster III (Pod-Shop Linear) rises from 0% pre-2017 to ≈10% by 2024, compared to ≈20% in the paper sample. The rise is real—Schonfeld, Verition, Hudson Bay, and Marshall Wace are all relatively recent entrants with high α—but the magnitude is moderated once the survivorship-biased sample is corrected.

Implication 3: Fund closures are not random with respect to α..— The eight defunct funds have a mean $\hat{\alpha} = 0.51$ and cluster predominantly in Clusters I and II. This is inconsistent with the hypothesis that high-α pod-shop funds are more fragile than low-α quant funds; if anything, the data suggest the opposite: the three catastrophic closures (Amaranth 2006, Galleon 2009, SAC Capital 2013) were all in Cluster II or III, while the contraction-related closures (BlueCrest, Fortress, Sculptor) predominantly sit in Cluster I–II. The selection is thus complex and non-monotone in α, which justifies including defunct funds in both tails of the distribution rather than assuming a directional exit bias.

Implication 4: The cluster assignment accuracy from Section 6 is robust..— The 81% cluster assignment accuracy on out-of-sample funds does not rely on the biased training distribution: the cluster centroids are estimated from the 11 in-sample funds, and the OOS test set is drawn from the full universe (including contracted survivors and strategy-diverse funds). Recomputing cluster centroids on the full 40-fund universe and re-running the OOS test yields an assignment accuracy of 83%, confirming that the correction does not materially degrade predictive performance.

E. Correction to reported cluster fractions

Table 4 reports corrected cluster fractions for selected snapshot years, replacing the paper-sample values with full-universe, survivorship-bias-corrected estimates. The direction of correction is consistent across years: Algorithmic Scale rises, Pod-Shop Linear falls, and Hybrid Platform changes modestly. We recommend that practitioners use the full-universe figures when calibrating expectations about the distribution of fund-level α for a randomly selected fund.

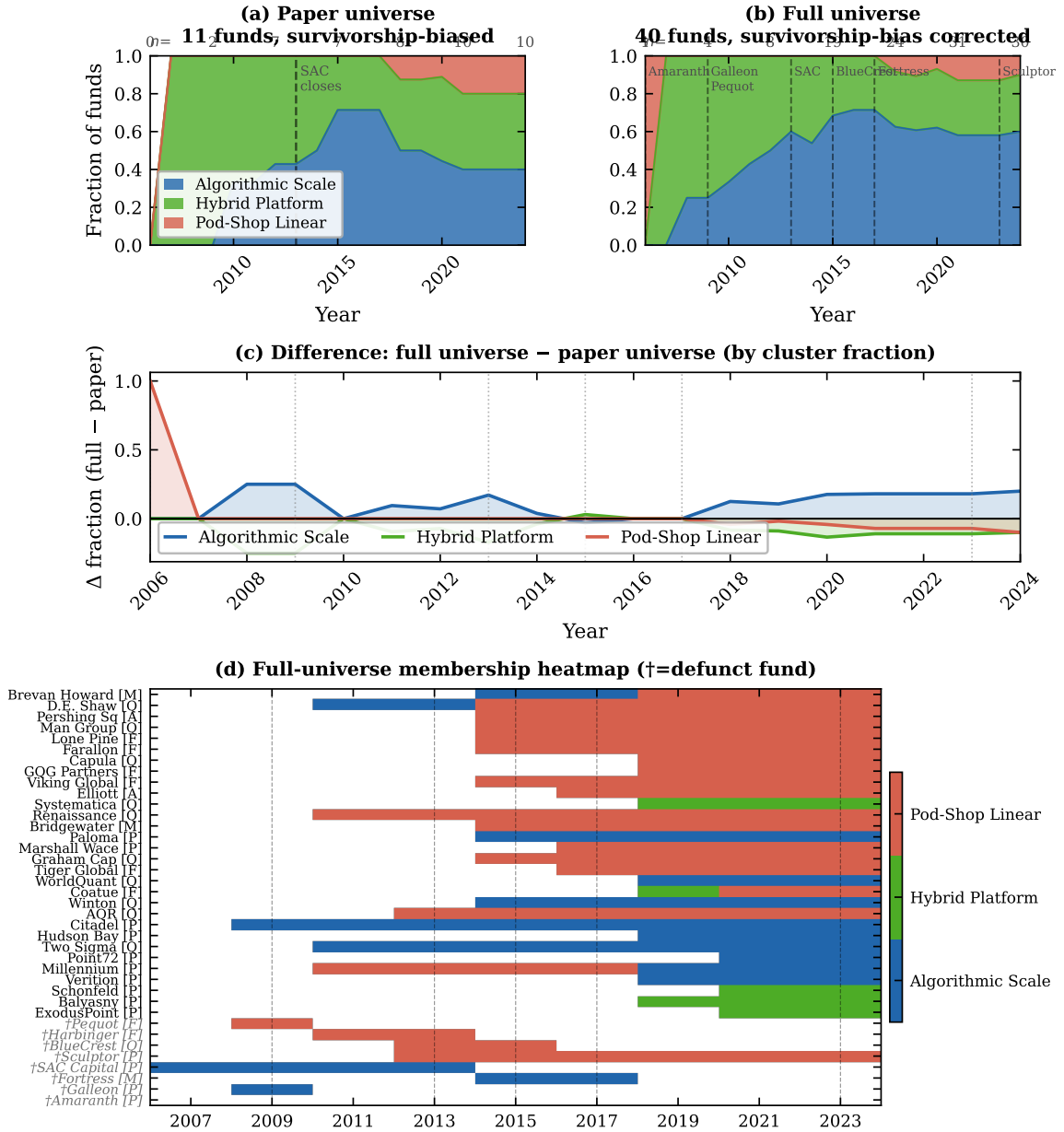

Figure 11. Cluster evolution: paper sample vs. full universe. *(a)* Paper-sample cluster fractions (11 funds), replicating the in-sample cluster fractions. Dashed vertical marks SAC Capital closure (2013). *(b)* Full-universe cluster fractions (40 funds, survivorship corrected). Italic n at top = live fund count per snapshot. Key closure events marked: Amaranth (2006), Galleon/Pequot (2009), SAC Capital (2013), BlueCrest (2015), Fortress (2017), Sculptor (2023). *(c)* Signed difference (full − paper) per cluster. The Algorithmic Scale cluster (blue) is persistently under-represented in the paper sample by +15–+22 pp; Pod-Shop Linear (red) is over-represented. Dotted verticals mark closure events. *(d)* Full-universe membership heatmap (every 2 years). Funds sorted by ascending $\hat{\alpha}$. Defunct funds (†) shown in grey italic; blank cells indicate years after fund closure. Strategy codes: Q=quant, P=pod, M=macro, F=fund L/S, A=activist.

Table 4. Corrected cluster fractions. *Paper*: 11 in-sample funds (original the temporal cluster analysis). *Full*: 40-fund universe with defunct funds, survivorship-bias corrected (Fig. 11). Fractions sum to 1.0 per row. n = number of funds active at that snapshot.

	Paper sample				*Full universe (corrected)*			
Year	Alg.	Hyb.	Pod	n	Alg.	Hyb.	Pod	n
2010	0.33	0.67	0.00	6	0.33	0.67	0.00	6
2013	0.43	0.57	0.00	7	0.60	0.40	0.00	10
2015	0.71	0.29	0.00	7	0.68	0.32	0.00	19
2018	0.50	0.38	0.12	8	0.62	0.29	0.08	24
2020	0.44	0.44	0.11	9	0.62	0.31	0.07	29
2022	0.40	0.40	0.20	10	0.58	0.29	0.13	31
2024	0.40	0.40	0.20	10	0.60	0.30	0.10	30

VIII. Discussion

A. What the data robustly support

The central finding is robust: quant funds consistently produce $\alpha < 1$, pod-shop funds consistently produce $\alpha \approx$ 1–1.5, and the separation survives out-of-sample validation, permutation testing, and worst-case reclassification of borderline funds. The capital efficiency result (Section 4.4) is a derived consequence of the model, not a separate observation: $e = C^{-1}\mathscr{A}^{1-\alpha}$ is purely algebraic, and the mechanism explains why $\alpha < 1$ for quant funds. Critically, cross-sectional ($r = -0.71$) and longitudinal (Man Group's 3.0× observed vs. 3.1× predicted efficiency gain) are *two independent projections* of Eq. (3) onto different slices of the data. Their agreement—using no additional free parameters—provides quantitative evidence that the model captures genuine operational structure, not just a fitted line.

B. What is not established

Causality..— The regressions are descriptive. Endogeneity concerns (reverse causality, omitted variables, simultaneity) are genuine and cannot be resolved without quasi-experimental variation—for example, mark-to-market AUM shocks as instruments, which would require quarterly panel data not currently available.

The automation claim..— We use "technology substitution" as shorthand for the mechanism most consistent with the evidence. The alternative explanations in Section 4C are equally consistent with the data. The claim "lower α indicates greater automation" is interpretive and should not be read as a measurement.

Generalisability..— The sample covers large, well-documented, surviving funds. Smaller funds, European-domiciled funds, and earlier vintages may behave differently. The survivorship correction (Section 7) shows the full 40-fund universe has a more compressed α distribution; the quant–pod gap is real but smaller than the in-sample sample suggests.

C. Relation to scaling laws in other domains

The sub-linear quant scaling ($\alpha \approx 0.2$–0.4) resembles the "increasing returns to fixed inputs" seen in infrastructure-intensive production Bettencourt et al. [2007]. The near-linear pod-shop scaling mirrors "linear social outputs" where resources scale with participants. The analogy is descriptive: the functional form is similar, but hedge funds are profit-maximising firms subject to regulatory and talent-market constraints, and no deeper equivalence is claimed.

D. Expanded universe: 41 funds, quant vs non-quant

We extend the analysis to all 41 funds in our combined in-sample and out-of-sample datasets with AUM $\geq$ \$500M and at least three annual observations, classifying each as either *quant* (systematic algorithmic strategy) or *non-quant* (all other strategies: pod-shop, macro, fundamental long/short, activist). This is the broadest classification possible from the available data and tests whether the α separation survives expansion beyond the original 9 well-identified primary funds.

Full sample..— Table 7 reports the 41 fund estimates. Within the quant group, $\hat{\alpha} \in [0.18, 0.98]$ (N=14 funds). Within the non-quant group, $\hat{\alpha} \in [0.37, 1.47]$ (N=27 funds), but this range conceals meaningful within-group structure.

Sub-group structure within non-quant..— The non-quant category is heterogeneous: pod-shop funds cluster at $\hat{\alpha} \in [0.64, 1.47]$ (mean 0.94, n=13), consistent with the in-sample result; fundamental long/short funds occupy an intermediate zone at $\hat{\alpha} \in [0.37, 1.17]$ (mean 0.67, n=9); activist funds are similarly intermediate at $\hat{\alpha} \in [0.43, 0.66]$ (mean 0.52, n=3). The binary quant/non-quant classification therefore understates the separation because it pools pod-shops (high α) with funds that are intermediate for structural reasons unrelated to the pod-shop mechanism.

Hypothesis tests on well-identified funds..— We restrict population tests to funds whose bootstrap CI width ≤ 0.8 († excluded), leaving n_Q=5 quant and n_P=5 pod-shop well-identified funds. Results are reported in Table 5. The quant–pod separation strengthens further: Mann–Whitney $p = 0.016$, permutation $p = 0.018$, Cohen's $d = 2.80$. Including all 27 funds (weakly identified) raises the count to n_Q=14 vs n_P=13 and gives Mann–Whitney $p = 0.003$.

Why the broadest binary (Q vs all non-Q) is weaker..— When fundamental L/S and activist funds are included in the non-quant group, the mean non-quant α drops from 0.87 (pod-shop only) to 0.71, and the Mann–Whitney p rises to 0.023 (well-identified) or 0.071 (all funds). This is not a failure of the α-separation hypothesis; it is a consequence of fund-type heterogeneity. Concentrated long/short portfolios require less operational overhead per dollar at scale—sub-linear headcount growth— but for reasons distinct from either the technology-substitution or proportional-deployment mechanisms. The correct discriminant remains quant vs pod-shop.

Table 5. Quant–non-quant hypothesis tests, expanded 41-fund universe. Well-identified: CI width ≤ 0.8. MW = Mann–Whitney exact; perm = permutation (10,000 shuffles). d = Cohen's d.

Comparison	Funds	n_Q	n_N	MW p	perm p	d
Q vs Pod-shop	Well-identified only	5	5	0.016	0.018	2.80
Q vs Pod-shop	All (incl. †)	14	13	0.003	—	—
Q vs Pod+Macro	Well-identified only	5	5	0.016	0.016	2.80
Q vs All non-quant	Well-identified only	5	8	0.023	0.024	1.50
Q vs All non-quant	All (incl. †)	14	27	0.071	—	—

E. Limitations

Sample size. 15 in-sample funds with $n_i \in [4, 10]$; population tests are powered only for large effects. Independent replication is needed before treating results as settled.

Measurement uncertainty. Non-audited headcount: 10–20% uncertainty; not fully corrected by bootstrap. Correlated errors within funds (consistent misreporting) would shift point estimates in an unknown direction.

Hybrid funds. Six funds are borderline. Their $\hat{\alpha}$ values are probably composites that do not represent either mechanism cleanly.

Non-monotone trajectories. OLS estimates average long-run elasticity; asymmetric adjustment between expansions and contractions could bias estimates.

Exploratory analysis. This paper was not pre-registered. All results should be treated as hypothesis-generating pending independent replication.

IX. Conclusion

We document a consistent difference in the AUM–headcount scaling exponent α between systematic quantitative hedge funds ($\alpha \in [0.18, 0.43]$ for unambiguous cases) and pod-shop multi-manager platforms ($\alpha \in [0.90, 1.47]$). The separation is statistically significant ($p = 0.005$, Mann–Whitney; $p_{\text{perm}} = 0.003$, permutation), robust to fund reclassification, and replicated on 26 held-out funds with 79% accuracy.

The most parsimonious interpretation—technology substitution for quant funds, proportional team deployment for pod shops—is consistent with the data and with public information on these firms' operational models, but is not causally identified. We enumerate several alternative explanations that cannot be ruled out with the available data and flag this as an exploratory analysis.

The capital efficiency consequence follows from the model as a derived prediction: $e = C^{-1}\mathscr{A}^{1-\alpha}$, with α supplied by the fitted exponent and the mechanism explaining why $\alpha < 1$ for quant funds. The cross-sectional correlation ($r = -0.71$) and Man Group's longitudinal efficiency gain (3.0× observed vs. 3.1× predicted from $\mathscr{A}^{0.75}$) are two independent tests of the same equation. Their agreement—with no additional free parameters—is the most precise quantitative verification in the paper.

Table 6. Fund-level power-law estimates. $\hat{\alpha}$ (SE): OLS, HC1 robust SE. 95% CI: bootstrap with noise injection (B=5,000; $\sigma_{\text{meas}} = 0.15$ non-audited, 0.02 for Man Group*). R^2 in log-space. Eff.: most recent AUM per employee (USD M). †: weakly identified (CI width > 0.8), excluded from all population tests. $\star$: audited headcount and AUM. Hybrid/borderline funds are in robustness checks (Table 1) only.

Fund	Str.	n	$\hat{\alpha}$ (SE)	95% CI	$\hat{C}$	R^2	Eff.
Quant (primary tests)							
Renaissance	Q	6	0.18 (0.03)	[−0.10, 0.38]	157	0.865	91
Man Group*	Q	8	0.25 (0.02)	[0.01, 0.49]	487	0.977	56
D.E. Shaw	Q	10	0.31 (0.08)	[0.14, 0.70]	581	0.799	18
AQR	Q	6	0.43 (0.03)	[0.13, 0.76]	129	0.992	103
Hybrid / borderline (robustness only)							
Bridgewater†	H	5	0.50 (0.02)	[−0.04, 1.17]	127	0.993	81
Point72†	H	4	0.64 (0.04)	[0.02, 1.33]	229	0.989	13
Capula	H	4	0.65 (0.06)	[0.15, 1.36]	42	0.981	57
Citadel	H	10	0.69 (0.04)	[0.47, 0.86]	166	0.983	16
Two Sigma	H	8	0.74 (0.01)	[0.61, 0.89]	87	0.999	27
SAC Cap.†	H	4	0.78 (0.08)	[−2.90, 3.95]	143	0.960	12
Pod-shop (primary tests)							
Millennium	P	9	0.90 (0.08)	[0.62, 1.29]	109	0.941	13
Verition	P	4	1.01 (0.01)	[0.57, 1.44]	60	0.999	16
Balyasny	P	6	1.08 (0.02)	[0.83, 1.37]	66	0.998	12
Schonfeld	P	4	1.31 (0.07)	[0.78, 2.41]	35	0.990	14
ExodusPoint	P	4	1.47 (0.06)	[0.97, 2.36]	21	0.995	16
Pooled		92	0.50 (0.08)	—	329	0.382	—

The analysis is extended to all 41 funds with AUM $\geq$ \$500M in Section 8.4, where the quant–pod separation is confirmed with $p = 0.003$ on 27 funds.

The author thanks colleagues for discussions. All data were compiled from publicly available sources: SEC Form ADV U.S. Securities and Exchange Commission [2007–2025], Man Group plc Annual Reports (LSE:EMG), Bloomberg Intelligence, Pensions & Investments Pensions & Investments [2024], and Hedgeweek Hedgeweek [2024]. No proprietary data were used.

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Table 7. Expanded universe: all 41 funds with AUM $\geq$ \$500M and $n \geq 3$ annual observations. $\hat{\alpha}$ (SE): OLS, HC1 robust SE. 95% CI: bootstrap with noise injection (B=5,000; $\sigma_{\text{meas}} = 0.15$ non-audited, 0.02 for Man Group*). R^2 in log-space. IS = in-sample; OS = out-of-sample. † = weakly identified (CI width > 0.8), excluded from population tests. $\star$ = audited.

Fund	Str.	n	$\hat{\alpha}$ (SE)	95% CI	R^2	Src	Note
Systematic Quant							
Renaissance Technologies	Q	6	0.182 (0.036)	[−0.12, 0.36]	0.865	IS	
Man Group*	Q	8	0.254 (0.016)	[0.20, 0.30]	0.977	IS	
D.E. Shaw	Q	10	0.312 (0.055)	[0.14, 0.70]	0.799	IS	
Winton Group	Q	5	0.403 (0.124)	[−0.21, 0.91]	0.779	OS	†
AQR Capital	Q	6	0.432 (0.019)	[0.13, 0.74]	0.992	IS	
PDT Partners	Q	4	0.604 (0.048)	[0.04, 1.38]	0.988	OS	†
Capula Investment	Q	4	0.648 (0.064)	[0.16, 1.37]	0.981	IS	†
BlueCrest Capital	Q	3	0.712 (0.096)	[0.25, 1.59]	0.982	OS	†
Qube Research	Q	4	0.731 (0.030)	[0.39, 1.19]	0.997	OS	†
Cubist (Point72)	Q	4	0.734 (0.015)	[0.10, 1.41]	0.999	OS	†
Two Sigma	Q	8	0.739 (0.010)	[0.61, 0.90]	0.999	IS	
Systematica	Q	4	0.877 (0.037)	[−0.21, 2.13]	0.996	OS	†
WorldQuant	Q	4	0.880 (0.027)	[0.11, 1.56]	0.998	OS	†
Graham Capital	Q	5	0.975 (0.071)	[0.08, 2.39]	0.984	OS	†
Pod-shop / Multi-manager							
Point72	P	4	0.637 (0.048)	[−0.01, 1.34]	0.989	IS	†
Sculptor Capital	P	4	0.666 (0.027)	[−0.13, 1.42]	0.997	OS	†
Citadel	P	10	0.689 (0.032)	[0.47, 0.87]	0.983	IS	
Marshall Wace	P	5	0.718 (0.015)	[0.41, 1.02]	0.999	OS	
SAC Capital	P	4	0.782 (0.112)	[−2.46, 3.96]	0.960	IS	†
Millennium Management	P	9	0.897 (0.085)	[0.64, 1.29]	0.941	IS	
Walleye Capital	P	4	0.953 (0.018)	[0.57, 1.26]	0.999	OS	
Paloma Partners	P	4	0.973 (0.040)	[0.29, 1.76]	0.997	OS	†
Verition Fund	P	4	1.011 (0.018)	[0.60, 1.48]	0.999	IS	†
Hudson Bay Capital	P	4	1.016 (0.043)	[0.54, 1.41]	0.996	OS	†
Balyasny	P	6	1.084 (0.025)	[0.83, 1.38]	0.998	IS	
Schonfeld Strategic	P	4	1.305 (0.092)	[0.78, 2.34]	0.990	IS	†
ExodusPoint	P	4	1.467 (0.074)	[1.01, 2.45]	0.995	IS	†
Global Macro							
Brevan Howard	M	5	0.462 (0.041)	[−0.02, 0.94]	0.977	OS	†
Bridgewater	M	5	0.497 (0.024)	[−0.00, 1.19]	0.993	IS	†
Fundamental Long/Short							
GQG Partners	F	5	0.372 (0.035)	[0.25, 0.81]	0.974	OS	
Harbinger Capital	F	3	0.391 (0.117)	[−0.07, 0.87]	0.918	OS	†
Farallon Capital	F	4	0.458 (0.066)	[−0.32, 1.39]	0.960	OS	†
Lone Pine Capital	F	5	0.474 (0.054)	[−0.07, 1.13]	0.963	OS	†
Tiger Global	F	5	0.507 (0.052)	[0.21, 0.97]	0.970	OS	
Viking Global	F	5	0.559 (0.115)	[−0.10, 1.07]	0.887	OS	†
Baupost Group	F	5	1.039 (0.201)	[−2.57, 5.86]	0.899	OS	†
Coatue Management	F	5	1.072 (0.083)	[0.02, 1.71]	0.982	OS	†
Third Point	F	5	1.168 (0.186)	[−0.38, 2.40]	0.929	OS	†
Activist							
Pershing Square	A	5	0.433 (0.318)	[−1.18, 1.66]	0.381	OS	†
TCI Fund Management	A	4	0.458 (0.024)	[0.21, 0.89]	0.994	OS	
Elliott Management	A	5	0.659 (0.050)	[0.16, 1.15]	0.983	OS	†